Temporally-structured multimodal prediction across six clinical targets in paediatric acute lymphoblastic leukaemia

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Abstract

Paediatric acute lymphoblastic leukaemia (ALL) is the most common childhood malignancy, yet 15–20% of patients relapse after apparently successful treatment. Existing risk-stratification approaches operate on single-timepoint snapshots and have not been systematically evaluated across the full clinical decision timeline nor validated across multiple independent centres. Here we present a temporally-structured multimodal prediction framework evaluated on six clinical targets simultaneously: four classification tasks (Final Risk, D19MRD, D46MRD, Relapse) and two survival endpoints (event-free survival, EFS; overall survival, OS), whose feature sets are strictly aligned with the clinical data-acquisition timeline from hospital admission to Day 46. Phase 1 evaluates four large language models (LLMs) in a zero-shot setting; Phase 2 trains decision trees, random forests, and random survival forests on a temporally partitioned case database. In the internal cohort, Phase 2 achieves AUROC 0.949 for Final Risk and concordance index 0.746 for EFS. A cosine similarity retrieval-augmented generation (RAG) belief revision improves D46MRD AUROC from 0.655 to 0.881 (**+0.226**) in the largest training regime. External validation at two independent sites confirms robust Final Risk transfer (LLM zero-shot AUROC 0.80–0.93; Phase 2 RF up to 1.000) without retraining, although the small external cohorts limit the remaining targets. This framework demonstrates that temporally stratified multimodal learning, aligned with clinical workflows, enables consistent multi-target prediction that generalises to independent cohorts.

Keywords: paediatric acute lymphoblastic leukaemia, temporal feature design, large language models, risk stratification, survival analysis, retrieval-augmented generation, external validation

1 Introduction

Acute lymphoblastic leukaemia (ALL) is the most common childhood malignancy, accounting for approximately 25% of all paediatric cancers [1]. Risk-adapted chemotherapy has elevated five-year event-free survival (EFS) rates above 90% in high-income settings, yet the 15–20% of patients who relapse face substantially worse outcomes, with long-term survival dropping to 30–50% upon re-induction [2, 3]. Because the majority of relapses occur after apparent clinical remission, the prospective identification of high-risk patients—early enough to intensify or de-escalate therapy—remains a central and unresolved challenge in paediatric oncology.

Current standard-of-care risk stratification—Initial Risk at diagnosis and Final Risk after the Day 46 induction assessment—relies on pre-specified categorical cut-offs for age, white blood cell count, cytogenetics, immunophenotype, and minimal residual disease (MRD). These rules do not fully exploit the multimodal information that accumulates during treatment, nor do they yield personalised probability estimates for future relapse or survival. Computational approaches promise to fill this gap, but most published models make simplifications that break down at the bedside: they treat outcome prediction as a static, single-shot problem, when in reality paediatric ALL management unfolds as a structured sequence of events in time.

We argue that realistic clinical prediction is governed by three distinct and largely independent timelines, and that existing methods collapse or ignore all three. The first is the *feature-acquisition timeline*: a patient's data do not arrive at once but in a fixed clinical order—white blood cell count and age at admission (Day 0), immunophenotype and bone marrow morphology (Day 1), fusion genes, karyotype and FISH (Day 7–10), and MRD (Day 19–46). A model trained on features pooled across all timepoints implicitly uses information unavailable at the moment of decision, producing optimistic but non-deployable estimates [9]. The second is the *prediction-target timeline*: clinicians do not make one prediction but a sequence of decisions—early response (D19MRD), end-of-induction response (D46MRD), Final Risk assignment, and long-horizon outcomes (relapse, EFS, overall survival [OS]). Studying any single target in isolation, as is common, obscures how one data-acquisition workflow must support all six decision nodes. The third is the *data-accumulation timeline*: a hospital's reference database grows year by year, so a model deployed in a given year can only have learned from patients enrolled before it. Random shuffled train–test splits violate this constraint and report performance that is not prospectively attainable. Aligning a dataset and a method to all three timelines simultaneously, rather than one at a time, is the central novelty of this work and the source of its clinical realism.

Building on this principle, we construct a dataset in which eight strictly nested feature sets ($\mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_8$) map onto the feature-acquisition timeline, six prediction targets map onto the target timeline, and patient enrolment year defines the data-accumulation timeline. On this substrate we evaluate two complementary methods. Phase 1 applies large language models (LLMs) in a zero-shot setting, probing the predictive upper bound reachable with *no* accumulated training data—the regime a clinician faces when a new protocol or a new hospital comes online [10]. Phase 2 trains decision trees, random forests, and random survival forests on the growing case database under temporal splits, emulating prospective deployment as the data-accumulation timeline advances. A retrieval-augmented generation (RAG) component then leverages the accumulated database to apply a similarity-weighted belief revision to model probabilities and to generate interpretable clinical narratives. Within the earliest feature sets, an automated bone marrow morphology pipeline (SAM-based cell segmentation and BiomedCLIP clustering) converts smear images into a structured cell-morphology profile, providing a multimodal enrichment at Day 1, before molecular and MRD data become available.

Our contributions are: (*i*) an eight-level, three-timeline feature-and-target hierarchy aligned with the clinical workflow, in which bone marrow morphology enters as an early multimodal enrichment; (*ii*) Phase 1, a systematic zero-shot evaluation of four LLMs across all feature levels and all six prediction targets, establishing training-free baselines; (*iii*) Phase 2, a decision tree, random forest, and random survival forest system trained on a dynamic case database partitioned by enrolment year and validated under three temporal splits; (*iv*) a cosine-similarity RAG belief revision that refines model probabilities and generates interpretable LLM clinical narratives; and (*v*) multi-site external validation at independent centres that received no training data.

2 Results

2.1 Study cohorts and temporally-structured feature design

The internal training cohort comprised paediatric ALL patients treated at [HOSPITAL NAME] between 2015 and 2019 (Table 1; $n = 301$). Prediction targets span the treatment timeline: Day 19 MRD positivity (D19MRD $\geq 1\%$), Day 46 MRD positivity (D46MRD $\geq 0.01\%$), final risk classification (Final Risk: LR vs IR/HR; 159 LR, 142 IR/HR), and relapse occurrence (44 of 301, 14.6%). EFS events (relapse, induction failure, or death; 75 of 301, 24.9%) and OS events (death; 24 of 301, 8.0%) together extend the prediction targets to six.

Two external validation cohorts from independent institutions were supplied as a single merged file, which we split by the recorded hospital of origin. External Site 1 (Lianyungang First People's Hospital) contributed 20 patients diagnosed in 2019–2023; External Site 2 (Children's Hospital of Fudan University, Shanghai) contributed 21 patients (2017–2021) (Table 1). The Site 2 cohort is relapse-enriched—all 21 patients relapsed—so at each site one or more targets reduce to a single class or a handful of events; these are reported with accuracy (classification) or flagged as indicative (survival) rather than dropped. For external validation, all 301 internal patients were used as the training set and no external data were used in training.

Feature sets $\mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_8$ (Table 5) are strictly nested: each level adds only features available at the corresponding clinical timepoint. $\mathcal{F}_1$ contains WBC count and age (Day 0); $\mathcal{F}_8$ additionally includes D19MRD and D46MRD values (Day 46). The feature hierarchy applies identically to all six prediction targets, with the distinction that classification tasks (targets 1–4) use AUROC as the primary metric, while survival targets (targets 5–6: EFS, OS) use Harrell's C-index and time-dependent IPCW AUC.

Table 1 Cohort characteristics and prediction target distributions. Ext-1: Lianyungang First People's Hospital; Ext-2: Children's Hospital of Fudan University (Shanghai). † patients with complete labels for all four classification targets. Note that the Ext-2 cohort is relapse-enriched: all 21 patients experienced relapse.

Characteristic	**Internal** ($n = 301$)	**Ext-1** ($n = 20$)	**Ext-2** ($n = 21$)
Enrolment years	2015–2019	2019–2023	2017–2021
B-ALL	163 (54.2%)	20 (100%)	18 (85.7%)
Patients with complete labels†	301	8	21
Classification targets			
Final Risk: LR	159 (52.8%)	11 (55.0%)	6 (28.6%)
Final Risk: IR/HR	142 (47.2%)	9 (45.0%)	15 (71.4%)
Relapse: Yes	44 (14.6%)	2 (10.0%)	21 (100%)
D19MRD $\geq$ 1%	[X] ([X]%)	0	4
D46MRD $\geq$ 0.01%	[X] ([X]%)	1	0
Survival endpoints			
EFS events	75 (24.9%)	3 (15.0%)	21 (100%)
OS events	24 (8.0%)	1 (5.0%)	6 (28.6%)
Multimodal data			
Bone marrow images	[X] (XX%)	20	21
Morphology features	287 (95.3%)	19	21

2.2 Phase 1: LLM zero-shot performance across six targets

We evaluated four LLMs in a zero-shot setting across all eight feature levels and all six prediction targets (Table 7 summarises the best level per model; Table 2 reports the complete per-feature-level results). Models included Gemini-3.1-Flash-Lite (Google AI API) and three open-weight models deployed locally via vLLM: Gemma-4-31B-it, InternVL3.5-30B-A3B, and Qwen3.6-35B-A3B. For classification tasks (targets 1–4), each patient's feature vector was serialised into natural language and the model output a categorical label; we report AUROC. For survival targets (5–6), the model output a continuous risk score (0–1) evaluated by concordance index (C-index).

Classification performance on the internal cohort. Final Risk was the easiest target for all models. At Day 0 features ($\mathcal{F}_1$: WBC and age), Gemini and Gemma already achieved AUROC $\approx$ 0.65–0.70, reflecting the known prognostic importance of WBC count. Performance increased incrementally and peaked at the cytogenetics panel $\mathcal{F}_7$ (Gemini 0.834, Gemma 0.891, InternVL 0.848, Qwen 0.882), which was the

Table 2 Phase 1 LLM zero-shot performance across all eight feature levels and six prediction targets (internal cohort, $n = 301$). Each cell is the macro-averaged AUROC (classification targets) or Harrell's concordance index (survival targets) for one LLM at one feature level; **bold** marks the best model at that level. Classification AUROC is computed from parsed categorical predictions (balanced-accuracy form). — indicates a target not evaluated at that level: D19/D46MRD are used as *input* features at $\mathcal{F}_8$ (see Table 5).

	Model					Model			
Lv.	Gem.	Gma.	Int.	Qw.	**Lv.**	Gem.	Gma.	Int.	Qw.
Final Risk (AUROC)					*Relapse (AUROC)*				
$\mathcal{F}_1$	0.788	0.785	0.729	**0.792**	$\mathcal{F}_1$	0.514	**0.572**	0.566	0.530
$\mathcal{F}_2$	0.782	**0.796**	0.539	0.789	$\mathcal{F}_2$	0.557	**0.564**	0.519	0.526
$\mathcal{F}_3$	0.812	**0.842**	0.583	0.831	$\mathcal{F}_3$	**0.577**	0.572	0.503	0.503
$\mathcal{F}_4$	0.776	**0.844**	0.571	0.818	$\mathcal{F}_4$	**0.584**	0.568	0.518	0.573
$\mathcal{F}_5$	0.826	0.835	0.591	**0.837**	$\mathcal{F}_5$	**0.575**	0.555	0.500	0.552
$\mathcal{F}_6$	0.791	**0.854**	0.823	0.845	$\mathcal{F}_6$	**0.596**	0.558	0.554	0.504
$\mathcal{F}_7$	0.834	**0.891**	0.848	0.882	$\mathcal{F}_7$	0.599	0.552	**0.643**	0.546
$\mathcal{F}_8$	0.849	0.788	0.716	**0.856**	$\mathcal{F}_8$	0.567	**0.586**	0.540	0.557
D19 MRD (AUROC)					*EFS (C-index)*				
$\mathcal{F}_1$	**0.571**	0.543	0.564	0.531	$\mathcal{F}_1$	0.577	**0.591**	0.584	0.573
$\mathcal{F}_2$	0.543	**0.555**	0.479	0.521	$\mathcal{F}_2$	**0.607**	0.598	0.522	0.561
$\mathcal{F}_3$	**0.631**	0.599	0.484	0.557	$\mathcal{F}_3$	0.587	**0.614**	0.545	0.537
$\mathcal{F}_4$	0.556	**0.659**	0.492	0.432	$\mathcal{F}_4$	0.597	**0.619**	0.532	0.533
$\mathcal{F}_5$	0.583	**0.597**	0.488	0.520	$\mathcal{F}_5$	0.598	**0.614**	0.526	0.561
$\mathcal{F}_6$	0.603	**0.620**	0.567	0.605	$\mathcal{F}_6$	**0.624**	0.583	0.604	0.542
$\mathcal{F}_7$	0.651	**0.652**	0.548	0.581	$\mathcal{F}_7$	**0.643**	0.630	0.621	0.581
$\mathcal{F}_8$	—	—	—	—	$\mathcal{F}_8$	0.687	**0.716**	0.642	0.662
D46 MRD (AUROC)					*OS (C-index)*				
$\mathcal{F}_1$	0.505	0.486	**0.542**	0.501	$\mathcal{F}_1$	**0.689**	0.648	0.652	0.661
$\mathcal{F}_2$	0.492	0.475	**0.501**	0.495	$\mathcal{F}_2$	0.647	**0.702**	0.550	0.634
$\mathcal{F}_3$	0.502	**0.509**	0.503	0.500	$\mathcal{F}_3$	**0.682**	0.678	0.537	0.634
$\mathcal{F}_4$	**0.528**	0.508	0.509	0.508	$\mathcal{F}_4$	0.667	**0.681**	0.524	0.616
$\mathcal{F}_5$	0.505	**0.539**	0.505	0.533	$\mathcal{F}_5$	0.687	**0.691**	0.497	0.660
$\mathcal{F}_6$	0.549	**0.577**	0.518	0.552	$\mathcal{F}_6$	0.701	0.704	**0.734**	0.619
$\mathcal{F}_7$	**0.610**	0.539	0.523	0.510	$\mathcal{F}_7$	0.667	0.693	**0.755**	0.687
$\mathcal{F}_8$	—	—	—	—	$\mathcal{F}_8$	0.725	0.718	**0.764**	0.718

best level for the three open-weight models; supplying the D19/D46 MRD values as inputs at $\mathcal{F}_8$ raised Gemini to 0.849 but did not further improve the others. Best-level Final Risk AUROCs were thus 0.849 (Gemini, $\mathcal{F}_8$), 0.891 (Gemma, $\mathcal{F}_7$), 0.848 (InternVL, $\mathcal{F}_7$), and 0.882 (Qwen, $\mathcal{F}_7$). Relapse prediction was substantially harder: best AUROCs were 0.643 (InternVL, $\mathcal{F}_7$), 0.599 (Gemini, $\mathcal{F}_7$), 0.586 (Gemma, $\mathcal{F}_8$), and 0.573 (Qwen, $\mathcal{F}_4$), with no consistent improvement as features accumulated. D19MRD and D46MRD showed moderate performance (AUROC 0.55–0.66), suggesting that LLMs can partially infer early treatment response from immunophenotypic and cytogenetic profiles but that these tasks remain harder without direct MRD measurement.

Survival performance on the internal cohort. For EFS, best-level C-indices were all attained at $\mathcal{F}_8$: 0.716 (Gemma), 0.687 (Gemini), 0.662 (Qwen), and 0.642 (InternVL). The inclusion of the D19/D46 MRD values at $\mathcal{F}_8$ was consistently optimal for EFS, confirming that residual disease burden encodes strong prognostic information accessible to LLMs without labelled training data. For OS—with only 24 events—C-indices also peaked at $\mathcal{F}_8$: InternVL reached 0.764, Gemini 0.725, and Gemma and Qwen both 0.718. Log-rank tests were significant ($p < 0.05$) at $\mathcal{F}_8$ for EFS for all four LLMs.

External validation (two sites). Final Risk—the only target with both classes at both sites—transferred well: LLM zero-shot AUROC was 0.80–0.90 at Site 1 (Lianyungang) and 0.80–0.93 at Site 2 (Fudan), close to the internal range and obtained without any site-specific retraining (Table 7). The other classification targets are limited by the small, class-imbalanced external cohorts: at each site one or more targets collapse to a single class (Relapse is all-positive at Site 2, D19MRD all-negative at Site 1), for which we report accuracy rather than AUROC. For survival, Site 2 (21 EFS events, 6 deaths) gave LLM EFS C-indices of 0.65–0.74 and OS 0.75–0.83; Site 1 estimates rest on very few events (3 EFS, 1 death) and are indicative only.

2.3 Bone marrow morphology pipeline

Bone marrow smear images were processed through a four-stage automated pipeline: SAM-based cell segmentation, morphological feature extraction ($\sim$37 features per cell), BiomedCLIP zero-shot cell-type clustering ($k = 7$ with Hungarian assignment), and per-patient aggregation into a quantitative morphology profile (Methods, Section 4.3). Morphology enters the feature timeline at Day 1 ($\mathcal{F}_4$), earlier than molecular and MRD data: the structured morphology profile is provided to the LLMs as the $\mathcal{F}_4$ input in Phase 1 and, in aggregated form, augments the tabular Phase 2 models. The latter gives the clearest benefit for early treatment-response prediction: adding the morphology features at $\mathcal{F}_4$ raised D19MRD to DT AUROC 0.743 / RF AUROC 0.807 (Section 2.4), showing that quantitative cell-level characterisation carries prognostic signal available already at diagnosis.

2.4 Phase 2: Temporal validation and classification+survival results

Phase 2 evaluates DT and RF models on three temporal splits ($\mathcal{T}_1 = 2016$: $n_{\text{train}} = 99$, $n_{\text{test}} = 201$; $\mathcal{T}_2 = 2017$: $n_{\text{train}} = 186$, $n_{\text{test}} = 114$; $\mathcal{T}_3 = 2018$: $n_{\text{train}} = 283$, $n_{\text{test}} = 17$) and a random survival forest (RSF) on the same splits for the EFS and OS targets (Table 7).

Classification on the internal cohort (Split 2). Final Risk DT AUROC rose from 0.810 at $\mathcal{F}_1$ to 0.912 at $\mathcal{F}_7$, with the largest single gain at $\mathcal{F}_6$ (+0.127, introduction of Initial Risk and Day 5 blast percentage). Including MRD at $\mathcal{F}_8$ further raised DT AUROC to **0.949** (RF: 0.998), the highest classification result across all conditions. D19MRD achieved DT AUROC 0.743 and RF AUROC 0.807 at $\mathcal{F}_4$, where the quantitative cell-morphology features from the pipeline augment the tabular model; D46MRD reached DT 0.671 and RF 0.636 at $\mathcal{F}_5$. Relapse prediction remained difficult for Phase 2 as for Phase 1 (best DT AUROC 0.529, RF 0.630), mirroring the broader literature on relapse biology [8].

Survival on the internal cohort (Split 2). RSF at $\mathcal{F}_8$ achieved C-index **0.746** and time-AUC at 24 months 0.775 for EFS (log-rank $p = 0.010$), outperforming the best LLM zero-shot EFS C-index (0.716) obtained without any training data. For OS, RSF C-index was 0.629 ($p = 0.489$; not significant), reflecting the low event rate (8.0%) and limited statistical power.

External validation (two sites, OOD). With all 301 internal patients as training, RF and DT retained strong Final Risk discrimination at both sites (RF AUROC 0.965 / DT 0.914 at Site 1; 1.000 / 1.000 at Site 2), confirming transfer to independent institutions. Consistent with Phase 1, the remaining per-site classification targets are constrained by class imbalance: several reduce to a single class (reported as accuracy) or rest on one to two positives, so their AUROCs are not interpretable (Table 7). For survival, the RSF transferred to Site 2 (EFS C-index 0.586 over 21 events; OS 0.705 over 6 deaths); the Site 1 RSF is omitted for OS (1 death) and rests on only 3 events for EFS.

2.5 Cosine RAG belief revision

For each test patient i, the $K = 5$ most similar training cases were retrieved by cosine similarity in the standardised feature space and used to compute a similarity-weighted RAG evidence probability p_{rag} (Eq. 4). The revised probability is $p_{\text{rev}} = \alpha\, p_0 + (1-\alpha)\, p_{\text{rag}}$, where $\alpha = 1 - (1-\alpha_{\text{base}})\sqrt{\bar{s}}$ and $\alpha_{\text{base}} = 0.6$.

Internal cohort. Table 3 reports AUROC before and after belief revision. Belief revision improved performance on Final Risk (Split 2: +0.002), D19MRD (Split 2: +0.012; Split 3: +0.024), and most substantially on D46MRD (Split 2: +0.046; Split 3: $0.655 \rightarrow 0.881$, $\Delta = +0.226$). The large D46MRD gain on Split 3 reflects high retrieval quality: with 283 training cases, the five nearest neighbours share strong phenotypic similarity (mean cosine $\bar{s} \approx 0.91$) and concordant D46MRD outcomes. Belief revision degraded Relapse performance (Split 2: $0.630 \rightarrow 0.588$, $\Delta = -0.042$): because relapse prediction is intrinsically noisy, retrieved neighbours have mixed outcomes, and RAG evidence acts as a noise source rather than a corrective signal.

External (pooled across sites). On the pooled external cohort, belief revision improved Final Risk ($0.991 \rightarrow 1.000$, +0.009). D19MRD AUROCs degraded (-0.064) because the external set has very few D19MRD-positive patients ($n_{\text{pos}} = 4$), making retrieved neighbour outcomes uninformative. D46MRD belief revision could not be computed ($n_{\text{pos}} = 1$). These results confirm the internal boundary condition: belief revision benefits tasks where the feature space clusters meaningfully by label, but is counter-productive when class representation is too sparse to yield informative neighbours.

Survival belief revision. For EFS and OS, belief revision was applied to normalised RSF risk scores using event indicators as binary outcomes. Effects were near-neutral for EFS (mean ΔC-index $= -0.003$ across splits) and slightly negative for

OS (mean ΔC-index = −0.007), where the low event rate (8%) renders most retrieved neighbours censored and the RAG evidence is uninformative.

Table 3 Cosine RAG belief revision results. p_{RF}: original RF AUROC; p_{rev}: AUROC after belief revision ($K = 5$, $\alpha_{\mathrm{base}} = 0.6$). Best experiment per task shown. † insufficient positives; belief revision not computed.

		Internal			External Site 1		
Task	**Split**	RF AUC	Rev AUC	Δ	RF AUC	Rev AUC	Δ
Final Risk	Split 2	0.998	0.999	+0.002	—	—	—
Final Risk	OOD	—	—	—	0.991	1.000	+0.009
D19MRD	Split 2	0.767	0.779	+0.012	—	—	—
D19MRD	Split 3	0.833	0.857	+0.024	—	—	—
D19MRD	OOD	—	—	—	0.335	0.270	−0.064
D46MRD	Split 2	0.533	0.579	+0.046	—	—	—
D46MRD	Split 3	0.655	**0.881**	**+0.226**	—	—	—
D46MRD	OOD	—	—	—		† ($n_{\mathrm{pos}} = 1$)	
Relapse	Split 2	0.630	0.588	−0.042	—	—	—
Relapse	OOD	—	—	—	0.267	0.283	+0.016

2.6 LLM clinical narratives and qualitative evaluation

After belief revision, an LLM prompt was constructed for each test patient containing: (i) the patient's serialised feature vector; (ii) the RF-predicted label, revised probability p_{rev}, and the human-readable DT decision path from root to leaf; (iii) the six patient-salient features ranked by $\mathrm{importance}_j \times |z_{ij}|$; and (iv) the structured profiles and outcomes of the top-5 retrieved historical cases. Gemini-3.1-Flash-Lite produced 3–5-sentence narratives for each prediction; all narratives from the $\mathcal{F}_8$/Split 3 experiment ($n = 17$ test patients) are available for physician review.

Two representative cases from the $\mathcal{F}_8$/Split 3 experiment illustrate the range of narrative quality (Fig. 6). In Case 1 (correctly classified IR, persistent MRD), the top-5 retrieved neighbours all share elevated MRD and cytogenetic abnormalities, and the narrative coherently attributes the high-risk classification to MRD non-clearance corroborated by historical precedent. In Case 2 (borderline case), the narrative highlights conflicting signals between MRD clearance and cytogenetic risk, suggesting next-generation sequencing as a way to resolve ambiguity.

Quantitative physician evaluation of narrative quality is planned and will follow the protocol in Table 4. Evaluation will cover: factual consistency with the patient's features, relevance of historical case comparisons, clinical plausibility of the interpretation, and overall helpfulness for clinical decision-making.

Table 4 Planned physician evaluation of LLM clinical narratives. Each narrative will be assessed on four dimensions by two independent clinicians using a 5-point Likert scale. Results to be filled following structured review.

Case ID	Factual accuracy (1–5)	Retrieval relevance (1–5)	Clinical plausibility (1–5)	Decision helpfulness (1–5)	Overal (1–5)
Case 001	—	—	—	—	—
Case 002	—	—	—	—	—
Case 003	—	—	—	—	—
⋮	⋮	⋮	⋮	⋮	⋮
Case 017	—	—	—	—	—
Mean (SD)	—	—	—	—	—
Inter-rater κ	—	—	—	—	—

3 Discussion

This study presents a temporally-structured multimodal framework evaluated jointly across six ALL prediction targets—four classification tasks and two survival endpoints—and validated at an independent external centre. Several findings warrant discussion.

MRD as the dominant signal across all targets. The largest single performance gain in Phase 2—DT AUROC for Final Risk from 0.912 ($\mathcal{F}_7$, cytogenetics) to 0.949 ($\mathcal{F}_8$, plus D19/D46 MRD)—quantitatively confirms the central role of MRD in clinical risk stratification [11]. The same pattern holds for LLM EFS C-index, which peaked at feature levels including MRD (Gemma: 0.716 at $\mathcal{F}_8$), and for Phase 2 RSF EFS (C-index 0.746, $p = 0.010$). These convergent results from four LLMs, decision trees, and a survival forest provide multi-method evidence that residual disease burden at Day 46 is the strongest prognostic signal accessible before relapse.

LLM zero-shot as a competitive baseline. Phase 1 Final Risk AUROCs of 0.848–0.891 without labelled training data are competitive with Phase 2 DT at $\mathcal{F}_7$ (0.912) and only modestly below the Phase 2 DT peak (0.949 at $\mathcal{F}_8$). This indicates that clinical knowledge encoded during LLM pretraining includes a meaningful understanding of paediatric ALL risk factors, and that LLMs may serve as strong priors when historical case databases are small or absent—for example, at the time a new treatment protocol is introduced. On external Site 1, LLM Final Risk AUROCs of 0.810–0.925 surpassed the internal zero-shot values for some models, likely due to differences in class distribution (larger IR/HR proportion in the external cohort) rather than genuine improvement in discriminative ability.

External validation: generalisation and limitations. Phase 2 RF Final Risk AUROC of 0.991 on External Site 1 (versus 0.998 internally) demonstrates strong cross-site generalisation. However, D19MRD AUROC dropped from 0.807 (internal) to 0.635 (external), indicating feature-distribution shift that partly limits generalisation for this target. The very low positive rate for D46MRD at the external site ($n = 1$) precluded belief revision and produced unstable AUC estimates, highlighting the need for larger external cohorts with balanced outcome distributions. Both external sites are small (20 and 21 patients) and the Site 2 referral cohort is relapse-enriched (all relapsed), so several per-site targets are single-class or event-sparse; larger, prospectively balanced external cohorts remain necessary for a definitive test of cross-site robustness.

Integrating six targets in a unified framework. Treating EFS and OS as targets 5–6 of the same feature hierarchy—rather than studying them in a separate survival analysis—reveals a coherent picture: the same feature level ($\mathcal{F}_8$) that maximises Final Risk classification also maximises EFS survival discrimination (RSF C-index 0.746), while Relapse and OS remain challenging regardless of feature set. This joint view avoids the selective reporting that can arise when survival and classification tasks are presented in separate papers or using different patient subsets.

Boundary conditions for RAG belief revision. Belief revision delivers quantitative gains specifically when the feature space clusters meaningfully by label. For D46MRD at Split 3 (mean similarity $\bar{s} \approx 0.91$), the $\Delta + 0.226$ AUROC gain is substantial and clinically relevant. For Relapse ($\Delta - 0.042$ internally, $+0.016$ externally with very low base AUROC), and for OS (mean ΔC-index $= -0.007$), retrieved neighbours are too heterogeneous to provide reliable evidence. A practical deployment criterion would be to apply belief revision only when inter-neighbour outcome agreement exceeds a calibrated threshold—for example, Fleiss $\kappa > 0.4$ over the top-K retrieved labels.

Relapse remains an open problem. Relapse prediction was intractable for all Phase 1 and Phase 2 models (best DT AUROC 0.529, LLM best 0.658), consistent with the broader literature [4, 8]. Our feature set does not include genomic or epigenomic data (e.g., copy-number variation, DNA methylation), which have shown modest incremental utility [6]. Future multi-omics integration within the temporal feature hierarchy may improve relapse prediction.

Limitations. Both cohorts are single-institution retrospective studies; multicentre prospective validation is required before clinical deployment. The internal cohort ($n = 301$) limits statistical power for OS (24 events) and Split 3 ($n_{\text{test}} = 17$). Phase 1 LLMs were evaluated zero-shot; fine-tuning on labelled clinical data may yield further improvements. Physician evaluation of LLM narratives (Table 4) is ongoing and will determine whether the narratives are clinically actionable beyond proof-of-concept case studies.

4 Methods

4.1 Datasets and ethics

The internal training cohort comprised $n = 301$ patients diagnosed with ALL at [HOSPITAL NAME] between 2015 and 2019. All patients were treated with risk-stratified chemotherapy protocols consistent with contemporary Chinese paediatric oncology guidelines. Ethics approval was obtained from [ETHICS NUMBER]; all data were de-identified prior to analysis. Each patient record includes structured clinical data (demographics, haematological indices, immunophenotype, 51-marker flow cytometry panel, cytogenetics, MRD values) and bone marrow aspirate microscopy images (Giemsa-stained, ∼20 fields per patient).

The external validation data were supplied as a single merged file spanning two independent institutions, distinguished by the recorded hospital field: External Site 1 (Lianyungang First People's Hospital; 20 patients, 2019–2023) and External Site 2 (Children's Hospital of Fudan University, Shanghai; 21 patients, 2017–2021, a relapse-enriched referral cohort in which all patients relapsed); ethics approval: [ETHICS NUMBERS Ext-1, Ext-2]. We evaluate the two sites separately. No external data were used in any model training.

4.2 Temporally-structured feature hierarchy

Feature sets form a strictly nested sequence $\mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_8$, where each level contains exactly the features available at the corresponding clinical timepoint. Boundaries: $\mathcal{F}_1$ (Day 0): WBC count and age; $\mathcal{F}_2$ (Day 1): adds blast percentage; $\mathcal{F}_3$ (Day 1): adds immunophenotype and 51 flow cytometry markers; $\mathcal{F}_4$ (Day 1): adds the bone marrow morphology profile (pipeline); $\mathcal{F}_5$ (Day 1): adds non-risk clinical factors; $\mathcal{F}_6$ (Day 5): adds CNS status, Day 5 blast, and Initial Risk; $\mathcal{F}_7$ (Day 7–10): adds Fusion gene, Karyotype, FISH; $\mathcal{F}_8$ (Day 46): adds D19MRD and D46MRD. The complete variable list for every level is given in Table 6. All six prediction targets are evaluated on the same feature hierarchy.

For Phase 2, categorical features were one-hot encoded and numerical features z-score normalised using training-split statistics only:

$$z_{ij}^{(\text{test})} = \frac{x_{ij}^{(\text{test})} - \mu_j^{(\text{train})}}{\sigma_j^{(\text{train})}}, \tag{1}$$

with missing values imputed by training-split medians.

4.3 Bone marrow morphology pipeline

Stage 1: Cell segmentation. SAM 3.1 [13] was applied with the text prompt *"complete dark-stained nucleated cell"* after colour-based pre-filtering (HSV hue 120–160°) to restrict sampling to haematoxylin-stained regions.

Stage 2: Morphological feature extraction. For each retained cell, ∼37 features were computed: geometric descriptors (area, perimeter, circularity, eccentricity, aspect ratio), textural descriptors (GLCM contrast and correlation), nucleus-to-cytoplasm (N/C) ratio, nucleoli count, and vacuole percentage.

Stage 3: Cell-type clustering. BiomedCLIP [14] was applied in a zero-shot clustering mode with $k = 7$ prototype classes (blast, granulocyte, megakaryocyte, monocyte, plasma cell, red cell, other), with Hungarian algorithm assignment.

Stage 4: Per-patient aggregation. Cell-level features were aggregated into per-patient morphology statistics (cell-type proportions and summary descriptors) that augment the tabular Phase 2 models; the same profile is serialised as the $\mathcal{F}_4$ input to the LLMs.

4.4 Phase 1: LLM zero-shot inference

For classification tasks, each patient's features were serialised into a key–value list and presented with a task-specific system prompt requesting a bracketed label (e.g., `[[IR]]`/`[[LR]]`). For survival risk scoring, the same serialised features were presented with a prompt requesting a continuous event probability (0.0–1.0). All experiments used temperature = 0 (greedy decoding). Per-patient outputs were cached; the cohort was scored for every combination of feature level (×12), task (×6), and model (×4).

4.5 Phase 2: Decision tree and random forest

A decision tree (DT; `max_depth`=4, `class_weight`='balanced', `criterion`='gini', `random_state`=42) and a random forest (RF; $n_{\text{estimators}} = 100$, `class_weight`='balanced') were trained for each feature level and temporal split. Three temporal splits defined by enrolment year threshold $\mathcal{T}_s$ (Eq. 2) ensure strict temporal non-leakage.

$$\mathcal{D}_s^{\text{train}} = \{i : \tau_i \leq \mathcal{T}_s\}, \quad \mathcal{D}_s^{\text{test}} = \{i : \tau_i > \mathcal{T}_s\}. \tag{2}$$

Primary metric: AUROC; secondary: F1-macro and balanced accuracy.

4.6 Phase 2: Random survival forest

A random survival forest (RSF; [17]; $n_{\text{estimators}} = 100$, `random_state`=42) was trained on the same temporal splits as the classification experiments. EFS was defined as time from diagnosis to first of: relapse, induction failure, or death; OS was time to death. Censoring was at last follow-up. Evaluation used Harrell's C-index (Eq. 6), IPCW time-dependent AUC (Eq. 7) at $t^* \in \{12, 24, 36\}$ months, and the log-rank test on Kaplan-Meier curves dichotomised at the median risk score.

4.7 External OOD evaluation

For external validation, all $n = 301$ internal patients served as the training set and external patients served as the test set; no site-specific fine-tuning was performed. DT and RF were trained on the full internal set using the same preprocessing pipeline (Eq. 1). RSF was trained identically. For Phase 1 LLM evaluation, no training was involved; the same zero-shot prompts were applied to external patient feature vectors.

4.8 RAG belief revision and LLM narration

For test patient i, the $K = 5$ most similar training cases were retrieved by cosine similarity in the standardised feature space:

$$\mathcal{N}(i) = \underset{j \in \mathcal{D}^{\text{train}}}{\arg\,\text{top-}K} \frac{\mathbf{z}_i \cdot \mathbf{z}_j}{\|\mathbf{z}_i\|_2 \|\mathbf{z}_j\|_2}, \quad K = 5. \tag{3}$$

The RAG evidence probability is

$$p_{\text{rag}} = \frac{\sum_{j \in \mathcal{N}(i)} s_j \, y_j}{\sum_{j \in \mathcal{N}(i)} s_j}, \tag{4}$$

where s_j is the cosine similarity to neighbour j and $y_j \in \{0, 1\}$ is the binary task outcome (or event indicator for survival). The revised probability is

$$p_{\text{rev}} = \alpha \, p_0 + (1 - \alpha) \, p_{\text{rag}}, \qquad \alpha = 1 - (1 - \alpha_{\text{base}})\sqrt{\bar{s}}, \tag{5}$$

where $\alpha_{\text{base}} = 0.6$ and $\bar{s} = \frac{1}{K} \sum_{j \in \mathcal{N}(i)} s_j$. For survival, p_0 is the min-max normalised RSF risk score.

After belief revision, an LLM prompt containing the patient's features, DT decision path, top-6 salient features (ranked by importance$_j \times |z_{ij}|$), and top-5 retrieved case profiles was submitted to Gemini-3.1-Flash-Lite to generate a 3–5 sentence clinical narrative.

4.9 Survival metrics

Harrell's concordance index:

$$\hat{C} = \frac{\sum_{i \neq j} \mathbf{1}[T_i < T_j] \, \mathbf{1}[\delta_i = 1] \, \mathbf{1}[\hat{r}_i > \hat{r}_j]}{\sum_{i \neq j} \mathbf{1}[T_i < T_j] \, \mathbf{1}[\delta_i = 1]}. \tag{6}$$

IPCW time-dependent AUC:

$$\widehat{\text{AUC}}(t^*) = \frac{\sum_{i:\, T_i \le t^*,\, \delta_i = 1} \sum_{j:\, T_j > t^*} \mathbf{1}[\hat{r}_i > \hat{r}_j] \, \hat{w}_i}{\sum_{i:\, T_i \le t^*,\, \delta_i = 1} \sum_{j:\, T_j > t^*} \hat{w}_i}, \tag{7}$$

where $\hat{w}_i = 1/\hat{G}(T_i)^2$ and $\hat{G}(\cdot)$ is the Kaplan-Meier censoring estimator [16].

4.10 Statistical analysis

AUROC was computed with `scikit-learn` [18]; C-index and IPCW AUC with `scikit-survival` [19]. Bootstrap 95% confidence intervals (1000 resamples) are reported for Split 3 ($n_{\text{test}} = 17$). No multiple-testing correction was applied given the exploratory nature of this study; all reported p-values are two-sided.

Data availability

The clinical dataset contains patient health information and cannot be made publicly available due to privacy regulations. De-identified data may be available through a data access agreement with [HOSPITAL NAME]; requests to the corresponding author.

Code availability

Analysis code, LLM prompts, and the bone marrow morphology pipeline will be made available at https://github.com/YanbeiJiang/ALL_prediction_NCS upon acceptance.

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Author contributions

[To be completed]

Competing interests

The authors declare no competing interests.

Table 5 Feature set design aligned with the clinical data-acquisition timeline. All six prediction targets are evaluated at every feature level; † at $\mathcal{F}_8$, D19MRD and D46MRD are used as *input features*, so only Final Risk and Relapse are predicted. EFS/OS are evaluated at all levels using the same feature vector as classification tasks.

Level	Timepoint	Cumulative features added	Class. targets
$\mathcal{F}_1$	Day 0	WBC count, Age	4
$\mathcal{F}_2$	Day 1	+ Blast%	4
$\mathcal{F}_3$	Day 1	+ Immunophenotype + 51 flow cytometry markers	4
$\mathcal{F}_4$	Day 1	$\mathcal{F}_3$ + bone marrow morphology profile	4
$\mathcal{F}_5$	Day 1	$\mathcal{F}_3$ + non-risk clinical factors	4
$\mathcal{F}_6$	Day 5	$\mathcal{F}_3$ + CNS status + Day 5 blast% + Initial Risk	4
$\mathcal{F}_7$	Day 7–10	$\mathcal{F}_6$ + Fusion gene + Karyotype + FISH	4
$\mathcal{F}_8$	Day 46	$\mathcal{F}_7$ + D19MRD + D46MRD (as inputs)	$2^{\dagger}$

Survival targets (EFS, OS) evaluated at all levels using the same feature vector.

Fig. 1 **Overview of the temporally-structured multimodal prediction framework. a,** The three parallel timelines: feature acquisition (Day 0–46, $\mathcal{F}_1$–$\mathcal{F}_8$), prediction targets (six tasks: Final Risk, D19MRD, D46MRD, Relapse, EFS, OS), and the dynamic case database (patient enrolment year 2015–2019, temporal splits). **b,** Phase 1 architecture: four LLMs receive serialised patient features and output class labels or risk scores without task-specific training. **c,** Phase 2 architecture: a decision tree trained on historical cases generates predictions and decision paths; RAG retrieves the $K = 5$ most similar cases; LLM synthesises a clinical narrative. **d,** Evaluation scope: internal temporal splits (three splits) and external OOD validation (two independent sites).

Table 6 Complete variable specification for each feature level (Supplementary Table). Each level adds the listed variables to its base set; $\mathcal{F}_4$ and $\mathcal{F}_5$ branch from $\mathcal{F}_3$, while the main clinical chain is $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_3 \subseteq \mathcal{F}_6 \subseteq \mathcal{F}_7 \subseteq \mathcal{F}_8$.

Level	Variables added (base set)
$\mathcal{F}_1$	WBC count; Age
$\mathcal{F}_2$	+ Blast%
$\mathcal{F}_3$	+ Immunophenotype (B/T lineage); 42 CD markers (CD1a, CD2, CD3, cCD3, CD4, CD5, CD7, CD8, CD9, CD10, CD11b, CD11c, CD13, CD14, CD15, CD19, CD20, CD22, cCD22, CD24, CD33, CD34, CD38, CD41, CD44, CD45, CD58, CD61, CD64, CD65, CD66c, CD71, CD73, cCD79a, CD86, CD90, CD97, CD99, CD117, CD123, CD133, CD200); 9 additional markers (GPA, cMPO, NG2, TdT, cμ, sIgM, TCR$\alpha\beta$, TCR$\gamma\delta$, HLA-DR)
$\mathcal{F}_4$	(base $\mathcal{F}_3$) + bone marrow morphology profile: per-patient blast-cell descriptors from the SAM/BiomedCLIP pipeline (area, perimeter, circularity, eccentricity, aspect ratio, nucleus-to-cytoplasm ratio, GLCM texture, chromatin fineness, nucleoli count/size, vacuole %, hand-mirror %) and cell-type proportions; Phase 2 uses the six SHAP-ranked descriptors
$\mathcal{F}_5$	(base $\mathcal{F}_3$) + non-risk clinical factors: chief-complaint category, pregnancy number, birth order, term/preterm status, delivery mode, birth weight, feeding type, admission weight, liver span, spleen span, haemoglobin, platelet count
$\mathcal{F}_6$	(base $\mathcal{F}_3$) + CNS status (CNSL); Day-5 peripheral blast%; Initial Risk
$\mathcal{F}_7$	(base $\mathcal{F}_6$) + Fusion gene; Karyotype; FISH
$\mathcal{F}_8$	(base $\mathcal{F}_7$) + D19MRD; D46MRD (as input features)

Fig. 2 Phase 1 feature-level ablation across six prediction targets. a, Final Risk AUROC vs feature level for four LLMs. **b,** Relapse AUROC. **c,** D19MRD AUROC. **d,** D46MRD AUROC. **e,** EFS C-index. **f,** OS C-index. Vertical dashed lines indicate clinical timepoints: Day 1 (flow cytometry and bone marrow morphology), Day 5 (early response), Day 7–10 (cytogenetics), Day 46 (MRD, $\mathcal{F}_8$). Open symbols: internal cohort. Filled symbols: External Site 1.

Fig. 3 Bone marrow morphology pipeline and its predictive contribution. a, Pipeline schematic: SAM cell segmentation → morphological feature extraction → BiomedCLIP cell-type clustering → per-patient morphology profile. **b,** Representative segmented cells with BiomedCLIP cell-type assignments. **c,** Predictive contribution of the morphology features in Phase 2: D19MRD DT/RF AUROC at $\mathcal{F}_3$ (no morphology) versus $\mathcal{F}_4$ (with morphology features).

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Table 7 Summary of best results across six prediction targets, four LLMs (Phase 1, best feature level), and Phase 2 models (DT/RF for classification; RSF for EFS/OS). Internal: Huaxi Hospital ($n = 301$; Phase 2 temporal Split 2). Ext-1: Lianyungang First People's Hospital ($n = 20$); Ext-2: Children's Hospital of Fudan University, Shanghai ($n = 21$), a relapse-enriched cohort (all 21 relapsed). Classification metric: AUROC; $*$ = accuracy, reported where the external cohort has a single class and AUROC is undefined. Survival metric: Harrell's C-index. Because the external cohorts are small, several cells rest on very few positives or events (e.g. Ext-1 Relapse: 2 positives; Ext-1 OS: 1 death, RSF omitted) and should be read as indicative only.

		Phase 1 LLM (best level)				**Phase 2**	
Task	**Cohort**	Gemini	Gemma	InternVL	Qwen	DT	RF/RSF
Classification tasks (AUROC; $*$ = accuracy, single-class cohort)							
Final Risk	Int.	0.849	0.891	0.848	0.882	0.949	0.998
	Ext-1	0.900	0.900	0.800	0.875	0.914	0.965
	Ext-2	0.933	0.883	0.800	0.867	1.000	1.000
D19MRD	Int.	0.651	0.659	0.567	0.605	0.743	0.807
	Ext-1	0.750*	0.750*	0.750*	0.875*	0.400*	1.000*
	Ext-2	0.515	0.449	0.500	0.537	0.581	0.750
D46MRD	Int.	0.610	0.577	0.542	0.552	0.671	0.636
	Ext-1	0.357	0.357	0.571	0.500	0.184	1.000
	Ext-2	0.714*	0.619*	0.571*	0.762*	0.667*	1.000*
Relapse	Int.	0.599	0.586	0.643	0.573	0.529	0.630
	Ext-1	0.750	0.750	0.750	0.750	0.431	0.861
	Ext-2	0.429*	0.524*	1.000*	0.191*	0.143*	0.048*
Survival tasks (C-index)							
EFS	Int.	0.687	0.716	0.642	0.662	—	0.746
	Ext-1	0.691	0.798	0.750	0.762	—	0.905
	Ext-2	0.741	0.712	0.677	0.653	—	0.586
OS	Int.	0.725	0.718	0.764	0.718	—	0.629
	Ext-1	1.000	1.000	0.972	0.972	—	—
	Ext-2	0.829	0.776	0.767	0.752	—	0.705

Fig. 4 Phase 2 temporal learning curves and classification results. a, Final Risk DT AUROC across feature levels for three temporal splits; shaded band: 95% bootstrap CI for Split 3 ($n_{\text{test}} = 17$). **b,** Training-set size vs AUROC for Final Risk and Relapse (DT and RF) at $\mathcal{F}_8$. **c,** External Site 1 RF AUROC comparison with internal Split 2 RF AUROC across all four classification tasks.

Fig. 5 Survival prediction: Kaplan-Meier curves and C-index summary. a, EFS Kaplan-Meier curves (high vs low risk, median split) for the best Phase 1 model (Gemma, $\mathcal{F}_8$, internal cohort). **b,** EFS KM curves for Phase 2 RSF (Split 2, $\mathcal{F}_8$, log-rank $p = 0.010$). **c,** OS KM curves for Phase 2 RSF on External Site 1 (log-rank $p = 0.031$). **d,** C-index heatmap: all LLMs $\times$ all feature levels for EFS (internal, left; Ext-1, right). Numbers at risk and 95% CI shown.

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Fig. 6 RAG belief revision and representative case narratives. a, AUROC before vs after belief revision across four tasks and three splits; point size proportional to training-set size. Filled points: internal; open points: External Site 1. **b–c,** Two representative cases from $\mathcal{F}_8$/Split 3. For each case: patient feature summary, DT decision path, top-3 retrieved similar historical cases with outcomes, and LLM-generated clinical narrative. Case b: correctly classified IR patient with persistent MRD and concordant historical neighbours. Case c: borderline case with conflicting MRD clearance vs cytogenetic risk signals.

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